Data Structures

CT077-3-2- DSTR(Version VE1)

Week 1

Introduction and Module Overview

Lecturer Information

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- Consultation Hours: Refer to iConsult

Pre-Requisites For This Module

- Fundamentals of Software Development (CT010-1-1)
- Or equivalent

Outcomes Based Education

- OBE is education based on producing particular educational outcomes that:
  - Focus on what students can actually do after they are taught.
  - Expect all learners / students to successfully achieve particular (sometimes minimum) level of knowledge and abilities.

- It’s NOT what We want to teach.
- It’s WHAT You should learn.

Aims of this Module

This module is designed to introduce common data structures and its related algorithms that are essential for solving computational problem.

The students will be exposed to implementation of such data structure using a suitable programming language.

Module Learning Outcomes

CLO	Learning Outcomes	Assessment
1	Explain various data structures and their applications (C2, PLO1)	Exam
2	Construct appropriate data structures to solve a given problem (C3, PLO2)	Assignment - Solution
3	Demonstrate a solid understanding of how to write solutions using various data structures and algorithms. (P5, PLO3)	Assignment - Lab Evaluation Work

Mapping of CLO with PLO

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO12
CLO1	✓											
CLO2		✓										
CLO3			✓									

The learning domains are:
PLO1: Knowledge and Understanding
PLO2: Cognitive Skills
PLO3: Practical Skills

Student Learning Time

- Module Credit Value: 3
- Total Learning Hours: 120 per semester

Lecture	28 hours
Tutorial	4 hours
Practical	24 hours
Others	0 hours
Independent Learning Time	42 hours
Assessment	22 hours
Total Learning Hours	120 hours per semester

Module Content Outline (Please add/reduce the column as needed)

Week	Topic
1	Overview on Data Structure
2 - 5	Fundamental Concept of Programming & Arrays
6 - 8	Linked Lists
9 - 10	Stacks
10	Queues
11 - 12	Trees
13 - 14	Graphs

Assessment Summary (refer to module handbook and module descriptor)

Form of Assessment	Assessment Methods	Hand Out Date	Hand In Date	%
Continuous Assessment	Lab Evaluation Works	6th week	14th week	45%
	Assignment – Solution (Report type submission)	6th week	10th week	15%
Final Assessment	Final Exam	16th week (Exam Weeks)	17th week	40%

Assessment requirement: Include any specific requirement to pass the module (refer to module handbook for the information), such as:

- To pass the module, you must attempt every element of assessment and achieve at least 50% in the module overall.

Expectations

1. Abide by ALL rules and regulations of APU.
2. Proper attire.
3. No speaking of dialects.
4. Attendance is compulsory. Valid Medical Certs must be supported in any absence from class.
5. Three cases of Late will be equal to 1 absence.
6. Use proper academic references – APA Referencing only.
7. Academic Dishonesty / Plagiarism is a serious offence. Any suspicions will be referred to the University's Academic Dishonesty Board.
8. Formal assessments must be submitted on time in the specified format given. Failure to meet deadlines will be treated as non-submission and no marks will be awarded. Incomplete submissions will be subjected to penalty of mark deductions or forfeit.

Other Expectations

- All laboratory assessment work (also known as lab evaluation works) must be completed during lab sessions. After the lab session, the absentee will not be allowed to submit the assignment and he/she will receive 0 points for that lab assessment.

- However, if the absentee holds a valid absence document, he/she can apply for Extenuating Circumstances (EC). The student will have the opportunity to resubmit to a lab course only if the Extenuating Circumstances (EC) request is approved.

- All laboratory assessment work is divided into group and individual components. The same group component score will be shared among all group members; however, individual components will receive different scores based on a member's single contribution to their answer. All group and individual scoring criteria will be clearly stated in the marking scheme later.

Achievement Requirements:

Undergraduate (Diploma, Foundation, Degree)

Marks	Alphabetical Grade	Grading Point	Classification
80-100	A+	4.0	Distinction
75-79	A	3.7	
70-74	B+	3.3	Credit
65-69	B	3.0	
60-64	C+	2.7	Pass
55-59	C	2.3	
50-54	C-	2.0	
40-49	D	1.7	Fail (marginal)
30-39	F+	1.3	Fail
20-29	F	1.0	Fail
0-19	F-	0	Fail

Reference Materials

Course Materials available in Moodle

- Module handbook
- Module descriptor
- Lecture slides
- Tutorial/Lab materials
- Sample incourse questions & answers
- Sample exam questions & answers

Essential and Further Readings

*Malik, D. S. (2017). C++ programming : program design including data structures. 8th ed. Clifton Park, NY : Cengage Learning. ISBN: 9781337117562

*Kedyk D. (2020). Implementing Useful Algorithms in C++. Independently published. ISBN: 9798605325307

Yang H. (2022). Easy Learning Data Structures & Algorithms C++: Explain Data Structures & Algorithms through full-color diagrams (C++ Foundation Design Patterns & Data Structures & Algorithms) . Independently published. ISBN: 9798436738994

*Note: Older references are indispensable and have no substitute as of Q1 2023

*Further readings will be assigned from time to time.

Your Valuable Feedback

- You are welcome to discuss your views on this module at any point of time.
- Do fill in anonymous evaluation questionnaires in the student feedback form. There are two points - mid and end of the teaching semester.
- The feedbacks you provide will be constructive for improvement of teaching and module content development.